24/06/2026

CODE-A
Phase-1

Aakash
Medical | IIT-JEE | Foundations

***Corporate Office*:** AESL, 3rd Floor, Incuspaze Campus-2, Plot no. 13, Sector-18, Udyog Vihar, Gurugram, Haryana-122015

FORTNIGHTLY TEST FOR NEET-2027 (XII PASSED)

MM: 720 **Test - 02** **Time: 180 Mins.**

Answers

1.	(1)	37.	(2)	73.	(4)	109.	(4)	145.	(3)
2.	(3)	38.	(4)	74.	(3)	110.	(1)	146.	(2)
3.	(4)	39.	(2)	75.	(1)	111.	(2)	147.	(4)
4.	(1)	40.	(4)	76.	(4)	112.	(4)	148.	(3)
5.	(4)	41.	(3)	77.	(3)	113.	(1)	149.	(4)
6.	(4)	42.	(2)	78.	(3)	114.	(2)	150.	(3)
7.	(4)	43.	(4)	79.	(4)	115.	(4)	151.	(4)
8.	(1)	44.	(3)	80.	(3)	116.	(4)	152.	(2)
9.	(2)	45.	(2)	81.	(2)	117.	(2)	153.	(4)
10.	(1)	46.	(2)	82.	(1)	118.	(1)	154.	(2)
11.	(3)	47.	(1)	83.	(2)	119.	(4)	155.	(2)
12.	(4)	48.	(3)	84.	(1)	120.	(4)	156.	(2)
13.	(3)	49.	(2)	85.	(1)	121.	(2)	157.	(2)
14.	(1)	50.	(1)	86.	(4)	122.	(4)	158.	(4)
15.	(4)	51.	(1)	87.	(1)	123.	(4)	159.	(3)
16.	(3)	52.	(4)	88.	(4)	124.	(2)	160.	(4)
17.	(4)	53.	(3)	89.	(3)	125.	(1)	161.	(4)
18.	(2)	54.	(2)	90.	(4)	126.	(2)	162.	(4)
19.	(2)	55.	(1)	91.	(3)	127.	(1)	163.	(4)
20.	(3)	56.	(3)	92.	(2)	128.	(1)	164.	(4)
21.	(1)	57.	(1)	93.	(1)	129.	(2)	165.	(2)
22.	(2)	58.	(4)	94.	(2)	130.	(3)	166.	(3)
23.	(3)	59.	(2)	95.	(4)	131.	(3)	167.	(4)
24.	(1)	60.	(4)	96.	(3)	132.	(3)	168.	(3)
25.	(4)	61.	(2)	97.	(2)	133.	(2)	169.	(1)
26.	(2)	62.	(2)	98.	(2)	134.	(4)	170.	(3)
27.	(1)	63.	(3)	99.	(1)	135.	(3)	171.	(3)
28.	(1)	64.	(2)	100.	(2)	136.	(2)	172.	(1)
29.	(3)	65.	(2)	101.	(4)	137.	(4)	173.	(1)
30.	(1)	66.	(1)	102.	(2)	138.	(1)	174.	(3)
31.	(4)	67.	(3)	103.	(4)	139.	(1)	175.	(1)
32.	(3)	68.	(1)	104.	(2)	140.	(3)	176.	(1)
33.	(2)	69.	(3)	105.	(1)	141.	(4)	177.	(2)
34.	(2)	70.	(2)	106.	(4)	142.	(2)	178.	(3)
35.	(3)	71.	(2)	107.	(3)	143.	(4)	179.	(2)
36.	(3)	72.	(4)	108.	(2)	144.	(3)	180.	(3)

Hints and Solutions

PHYSICS

(1) **Answer :** (1)
Solution:
Time to reach same level

$T_1 = \frac{2u\sin\theta}{g} = \frac{2\times 20\times \sin 30^\circ}{10} = 2\,\text{s}$

For complete flight

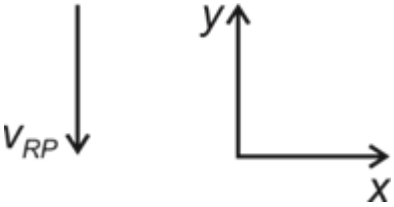
$y = u\sin\theta\, t_2 - \frac{1}{2}gt_2^2$

$-40 = 20\sin 30^\circ\, t_2 - 5\cdot t_2^2$

$t_2^2 - 2t_2 - 8 = 0,\ t_2 = 4\,\text{sec}$

$\frac{t_1}{t_2} = \frac{2}{4} = \frac{1}{2}$

(2) **Answer :** (3)
Hint:
$\vec{v}_{RP} = \vec{v}_R - \vec{v}_P$
Solution:



$\vec{v}_R = a\hat{i} - b\hat{j}$ $\qquad$ $\vec{v}_P = 4\hat{i}$

$\vec{v}_{RP} = \left(a-4\right)\hat{i} - b\hat{j}$

$a - 4 = 0$
$a = 4$
$b = 4$

$\left|\vec{v}_{RP}\right| = \sqrt{4^2+4^2}$

$= 4\sqrt{2}$ m/s

(3) **Answer :** (4)
Solution:

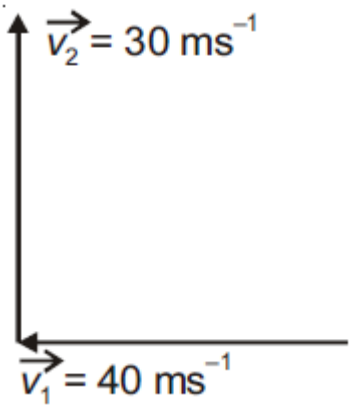
Using equation of trajectory $y = x\tan\theta\left(1-\frac{x}{R}\right)$

Where $R = \frac{u^2\sin 2\theta}{g} = \frac{12u^2}{125}$

$2.4 = 2.4\left(\frac{4}{3}\right)\left[1-\frac{2.4}{12u^2}(125)\right]$

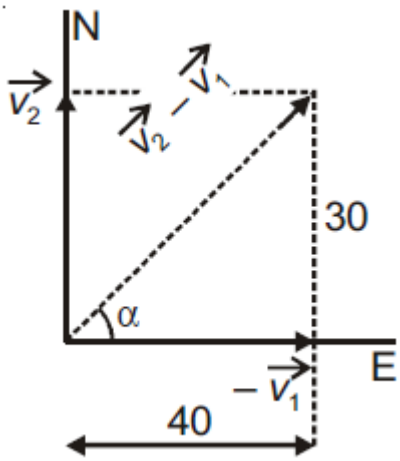
⇒ u = 10 m/s

(4) **Answer :** (1)
Solution:



Aakash Medical|IIT-JEE|Foundations

Average acceleration = $\frac{\vec{v}_2-\vec{v}_1}{\Delta t}$



$\tan\alpha = \frac{3}{4}$

α = 37°

(5) **Answer :** (4)
Hint:
Use vector addition.
Solution:
$\vec{P} = 3\hat{i}+4\hat{j}$
$\vec{P}+\vec{Q} = 8\hat{i}+7\hat{j}$
$\vec{Q} = 8\hat{i}+7\hat{j}-\vec{P} = 8\hat{i}+7\hat{j}-3\hat{i}-4\hat{j} = 5\hat{i}+3\hat{j}$
$\left|\vec{Q}\right| = \sqrt{25+9} = \sqrt{34}$

(6) **Answer :** (4)
Solution:
$R_{\text{res}} = |\hat{A}|+|\hat{B}|+|\hat{C}| = 1+1+1 = 3$

(7) **Answer :** (4)
Solution:
$x = 2t \Rightarrow t = \frac{x}{2}$
$y = 3t - \frac{1}{2}10t^2 = 3t - 5t^2$
$y = \frac{3x}{2} - 5\frac{x^2}{4} \Rightarrow 10x^2 - 12x + 8y = 0$

(8) **Answer :** (1)
Solution:
When a body moves with uniform velocity, its acceleration is zero.
$F_{\text{net}} = m \times a \Rightarrow$ If $a = 0$, then $F_{\text{net}} = 0$
Hence, body is said to be in equilibrium.

(9) **Answer :** (2)
Solution:
$T = \frac{2m_1m_2g}{m_1+m_2} = \frac{2\times3\times6\times10}{(3+6)} = 40$ N

(10) **Answer :** (1)
Solution:
$\tan\theta = \mu = \frac{dy}{dx}$
$y = 3x^2$
$\frac{dy}{dx} = 6x$ ... (i)
$\tan\theta = 0.75 = \frac{3}{4}$ ... (ii)
$\frac{3}{4} = 6x \Rightarrow x = \frac{1}{8}m$
$y = \frac{3}{64}m$

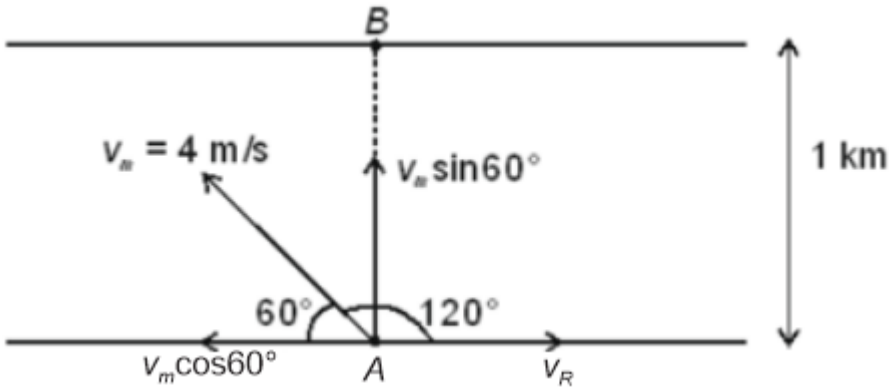
Aakash Medical|IIT-JEE|Foundations

(11) **Answer :** (3)
Hint:
Impulse = change in momentum
Solution:
$I = \Delta P = (10t^2 - 4t)$
$10t^2 - 4t = 0$
$t = 0, t = \frac{2}{5}$ s

(12) **Answer :** (4)
Solution:
Friction force also acts when there is tendency of relative motion. Hence statement I is incorrect. Raising the outer edge of a curved road above the inner edge to help vehicles negotiate the curve safely is called banking of the road.

(13) **Answer :** (3)
Solution:
$\mu mg = mr\omega^2$

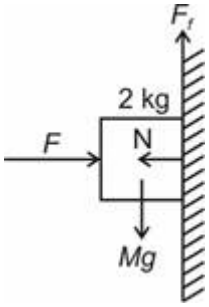
(14) **Answer :** (1)
Solution:



To reach at opposite point on other bank then net velocity along river must be zero.
v_m cos60° = v_R
$v_R = 4 \times \frac{1}{2} = 2\ m/s$
$\boxed{v_R = 2\ \text{m/s}}$

(15) **Answer :** (4)
Solution:
$v_R = -20\hat{j}$
$v_G = -10\hat{i}$
$\vec{v}_{R,G} = v_R - v_G$
$\vec{v}_{R,G} = 10\hat{i} - 20\hat{j}$
$\theta = \tan^{-1}\left(\frac{v_G}{v_R}\right) = \tan^{-1}\left(\frac{10}{20}\right) = \tan^{-1}\left(\frac{1}{2}\right)$
Towards west from vertical.

(16) **Answer :** (3)
Solution:



In horizontal equilibrium
$N - F = 0$
$N = F$
In vertical equilibrium
$\mu F = Mg$

$F = \frac{2\times 10}{0.4} = 50\ \text{N}$

(17) **Answer :** (4)
Solution:
$\vec{a}_{12} = \vec{a}_1 - \vec{a}_2$
$= -g\hat{j} - (-g\hat{i})$
= 0

(18) **Answer :** (2)
Hint:
check pulling force along inclined plane is more or less than limiting friction.
Solution:
Limiting friction F_L = μN = $\frac{\sqrt{3}}{2} \times 6 \times 10 \cos 30^\circ$
= 45 N
Force acting along the plane = *mg* sin 30°
$= 6 \times 10\frac{1}{2} = 30\ \text{N}$
$F_{\text{applied}} < f_L$
Hence *f* = 30 N

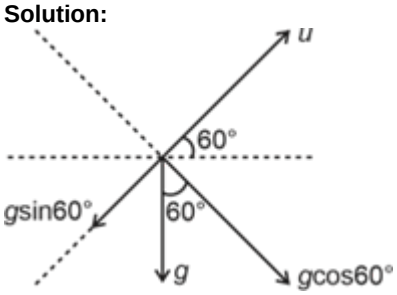
(19) **Answer :** (2)
Solution:
$10 - 6 \le \left|\vec{R}\right| \le 10 + 6$
$\Rightarrow 4 \le \left|\vec{R}\right| \le 16$

(20) **Answer :** (3)
Solution:
$2T\cos\theta = \sqrt{2}\,Mg$
$2\,Mg\cos\theta = \sqrt{2}\,Mg$
$\cos\theta = \frac{1}{\sqrt{2}}$
θ = 45°

(21) **Answer :** (1)
Solution:
Velocity $\left[\frac{\Delta x}{\Delta t} = \frac{2-0}{2-0} = 1\right]$ at *t* = 2 s changes from
1 m/s to –1 m/s
Impulse $j = \Delta p = 2mv = 2 \times 0.5 \times 1 = 1\ \text{N s}$

(22) **Answer :** (2)
Solution:
$\overrightarrow{v_{BA}} = \overrightarrow{v_B} - \overrightarrow{v_A} = 3\hat{i} + 5\hat{j} - 5\hat{i} + 2\hat{j} = -2\hat{i} + 7\hat{j}$

(23) **Answer :** (3)
Solution:



$|a_c| = g\cos 60^\circ = \frac{g}{2}$
$|a_t| = g\sin 60^\circ = \frac{g\sqrt{3}}{2}$
$\left|\frac{a_t}{a_c}\right| = \sqrt{3}$

(24) **Answer :** (1)
Solution:
Net contact force $= \sqrt{f^2 + N^2}$
$= \sqrt{(mg\sin\theta)^2 + (mg\cos\theta)^2}$

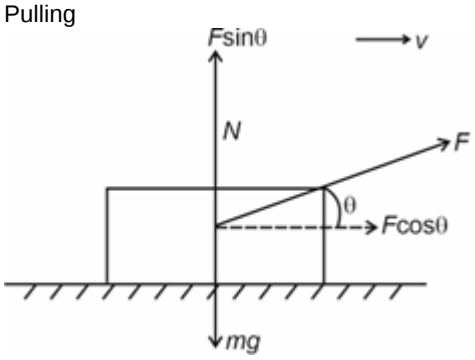
Aakash Medical|IIT-JEE|Foundations

$= mg\sqrt{\sin^2\theta + \cos^2\theta}$
= *mg*

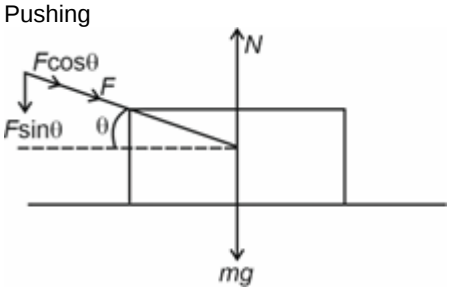
(25) Answer : (4)
Solution:
Action and reaction forces act on different bodies and cannot cancel each other.

(26) Answer : (2)
Hint:
Change in magnitude of velocity: $= 2u\sin\frac{\theta}{2}$
Step-1: $2u\sin\frac{180^\circ}{2} = 2u$

(27) Answer : (1)
Solution:
Pulling



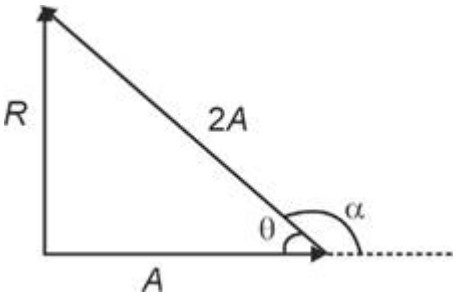
Along vertical
$\Rightarrow N + F\sin\theta = mg$
$\Rightarrow N = mg - F\sin\theta$
$\Rightarrow f_K = \mu_K(mg - F\sin\theta)$
Pushing



hence, $N = mg + F\sin\theta$ and $f_K = \mu_K N$
$\mu_K(mg + F\sin\theta)$
∴ It is clear in pulling friction is lesser than friction in pushing.

(28) Answer : (1)
Solution:
Friction coefficient of ice is small so to avoid slipping, one should take smaller step.

(29) Answer : (3)
Solution:



$\cos\theta = \frac{A}{2A} = \frac{1}{2} \Rightarrow \theta = 60^\circ$
∴ Angle between these two vectors = α = π – θ
= 180° – 60° = 120°

(30) Answer : (1)
Solution:
Since $T = 2\pi\sqrt{\frac{L\cos\theta}{g}}$

Aakash Medical|IIT-JEE|Foundations

$T^2 = \frac{(2\pi)L}{g}\cos\theta$

$\boxed{T^2 \propto \cos\theta}$

(31) **Answer :** (4)
Hint:
Friction force (*f*) = μ*N* = μ*mg*
Solution:
a = μ*g*
a = 0.3 × 10 = 3 m/s^2
v = *u* + *at*
v = 0 + 3 × 3 = 9 m/s

(32) **Answer :** (3)
Solution:
Third law of motion explains swimming.

(33) **Answer :** (2)
Solution:
$f = f\cos\theta$ block not moving

(34) **Answer :** (2)
Solution:
$F = v_g\frac{dm}{dt}$
$\frac{dm}{dt} = 50$ kg/s, $v_g = 3$ km-s^{-1}
$F = \left(3\times10^3\right)(50) = 15\times10^4$ N

(35) **Answer :** (3)
Solution:
Δp = 2 mu
F = Δp/Δt

(36) **Answer :** (3)
Solution:
$u = 6\hat{i} + 10\hat{j}$
$\vec{a} = 2\hat{i} - 3\hat{j}$
$V = u + at = \left(16\hat{i} + 10\hat{j}\right) + \left(2\hat{i} - 3\hat{j}\right)\times 2$
$= 6\hat{i} + 10\hat{j} + 4\hat{i} - 6\hat{j}$
$= 10\hat{i} + 4\hat{j}$
$|V| = \sqrt{10^2 + (4)^2} = \sqrt{116}$ ms^{-1}

Aakash Medical|IIT-JEE|Foundations

(37) **Answer :** (2)
Solution:
$p_3 = \sqrt{p_1^2 + p_2^2} = \sqrt{(1\times12)^2 + (2\times8)^2} = 20$
$5\times v_3 = 20$
v_3 = 4 m/s

(38) **Answer :** (4)
Solution:
Given $y = x - \frac{x^2}{80}$
On comparing with $y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$
$\tan\theta = 1 \Rightarrow \theta = 45^\circ$
Now, $\frac{g}{2u^2\cos^2\theta} = \frac{1}{80}$
$\Rightarrow \frac{10}{2u^2\times\left(\frac{1}{2}\right)} = \frac{1}{80}$
$u = \sqrt{800} = 20\sqrt{2}$ ms^{-1}
• Maximum height gained
$H_{\max} = \frac{u^2\sin^2\theta}{2g} = 20$ m
• Horizontal range
$R = \frac{u^2\sin 2\theta}{g} = 80$ m

(39) **Answer :** (2)
Hint:
$H = \frac{u^2 \sin^2 \theta}{2g}$
Solution:
$u \cos \theta = \frac{u}{2}$
$\cos \theta = \frac{1}{2}$
θ = 60°
$H = \frac{u^2 \sin^2 60^\circ}{2g}$
$H = \frac{3u^2}{8g}$

(40) **Answer :** (4)
Solution:
$\vec{F} = \frac{d\vec{F}}{dt}$
$\Rightarrow \quad \int d\overrightarrow{P} = \int_0^2 4t\, dt\, \hat{i} + \int_0^2 5dt\, \hat{j}$
$\Rightarrow \quad \Delta\overrightarrow{P} = 4\left[\frac{t^2}{2}\right]_0^2 \hat{i} + 5[t]_0^2 \hat{j}$
$\Rightarrow \quad \Delta\overrightarrow{P} = 8\hat{i} + 10\hat{j}$
$\Rightarrow \quad \left|\Delta\overrightarrow{P}\right| = \sqrt{64 - 100} = 2\sqrt{41}$
$\Rightarrow \quad m\left|\overrightarrow{\Delta v}\right| = 2\sqrt{41}$
$\Rightarrow \quad \left|\overrightarrow{\Delta v}\right| = \sqrt{41}$
$\because \quad v_{\text{initial}} = 0$
$\Rightarrow \quad v_{\text{final}} = \sqrt{41}$

(41) **Answer :** (3)
Solution:
u_x = 40 cos30° $= 20\sqrt{3}\ \text{m/s}$
distance AB = u_x ($t_2 - t_1$)
$= 20\sqrt{3} \times (3 - 1)$
$= 40\sqrt{3}\ \text{m}$

(42) **Answer :** (2)
Solution:
$y^2 + z^2 = a_0^2$
Which represents circular motion.

(43) **Answer :** (4)
Solution:
$T_2 = m(2) \times \omega^2 = 2m\omega^2$
$T_1 - T_2 = m(1)\omega^2$
$T_1 = T_2 + m(1)\omega^2 = 2m\omega^2 + m(1)\omega^2 = 3m\omega^2$
$\frac{T_1}{T_2} = \frac{3m\omega^2}{2m\omega^2} = \frac{3}{2}$

(44) **Answer :** (3)
Solution:
$F = (M + m)a$
$150 = 11 \times a$
$a = 13.6\ \text{m/s}^2$
For no relative motion between 10 kg and 1 kg
$1\ g = \mu \times ma$
$g = \mu \times a \Rightarrow a = \frac{10}{0.8} = 12.5\ \text{m/s}^2$
⇒ 10 kg and 1 kg will move together without falling down or moving up.

(45) **Answer :** (2)
Solution:

Aakash Medical|IIT-JEE|Foundations

$a_1 = \frac{3mg-mg}{m}$

$a_2 = \frac{(3m-m)}{(3m+m)}g$

CHEMISTRY

(46) Answer : (2)
Solution:
Order of (e/m) is electron > proton >α-Particle > neutron

(47) Answer : (1)
Solution:
Number of neutrons $= A - Z$
$^{13}_{6}C$ number of neutrons $= 13 - 6 = 7$
$^{14}_{7}N$ number of neutrons $= 14 - 7 = 7$

(48) Answer : (3)
Solution:
• Canal rays contain positively charged particles which are created by removal of electron from gaseous atom present in discharge tube.

(49) Answer : (2)
Solution:
The positive charge and most of the mass of the atom is concentrated in a small space called nucleus.

(50) Answer : (1)
Solution:
$\nu = \frac{C}{\lambda} = \frac{3\times10^8}{700\times10^{-9}} = 4.3\times10^{14}\ \text{Hertz}$

(51) Answer : (1)
Hint:
Photoelectric effect phenomenon can be explained by particle nature of electromagnetic radiation.
Solution:

Metal	W_0/eV
Li	2.42
Na	2.3

(52) Answer : (4)
Hint:
Power = rate of consumption of energy
Solution:
10 W = 10 Js^{-1}
Energy of photons = $\frac{hc}{\lambda} = \frac{6.626\times10^{-34}\times3\times10^8}{400\times10^{-9}}$
$= 4.9 \times 10^{-19}$
Rate of emission of photon
$= \frac{10}{4.9\times10^{-19}} = 2.04\times10^{19}$

(53) Answer : (3)
Solution:
$E = E_1 + E_2$
$\frac{hc}{\lambda} = \frac{hc}{\lambda_1} + \frac{hc}{\lambda_2}$
$\frac{1}{\lambda} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$
$\frac{1}{600} = \frac{1}{900} + \frac{1}{\lambda_2}$
$\therefore \lambda_2 = 1800$ nm

(54) Answer : (2)
Solution:

n	Size of orbital
l	Shape of orbital
m_l	Orientation of orbital

Aakash Medical|IIT-JEE|Foundations

m_S	Orientation of spin of electron

(55) Answer : (1)
Solution:
For Balmer series, $n_1 = 2,\ n_2 = 3, 4, 5, \ldots$
for shortest wavelength (λ_1), $n_1 = 2,\ n_2 = \infty$

$\therefore \frac{1}{\lambda_1} = RZ^2 \left[\frac{1}{{n_1}^2} - \frac{1}{{n_2}^2}\right]$

$\frac{1}{\lambda_1} = R \times (1)^2 \left[\frac{1}{2^2} - \frac{1}{\infty^2}\right] = R\left[\frac{1}{4} - 0\right]$

$\Rightarrow \lambda_1 = \frac{4}{R} \Rightarrow R = \frac{4}{\lambda_1}$

for Li^{2+} ion ($Z = 3$), longest wavelength in Paschen series, $n_1 = 3,\ n_2 = 4$

$\Rightarrow \frac{1}{\lambda_{Li^{2+}}} = R \times (Z)^2 \left[\frac{1}{3^2} - \frac{1}{4^2}\right]$

$\Rightarrow \frac{1}{\lambda_{Li^{2+}}} = R \times (Z)^2 \left[\frac{1}{9} - \frac{1}{16}\right]$

$\Rightarrow \frac{1}{\lambda_{Li^{2+}}} = R \times \left[1 - \frac{9}{16}\right] = \frac{7R}{16}$

$\Rightarrow \lambda_{\mathrm{Li}^{2+}} = \frac{4\lambda_1}{7}$

(56) Answer : (3)
Solution:
For H like species

$\cdot\ \mathrm{m_e vr} = \frac{\mathrm{n.h}}{2\pi}$

$\therefore$ for $\mathrm{n} = 3 \Rightarrow$ Angular momentum $= \frac{3}{2}\frac{\mathrm{h}}{\pi}$

• For H-atom radius of stationary state $r_n = n^2 a_0$

$\mathrm{r_1} = 1^2 \mathrm{a_0} = 52.9\ \mathrm{pm}$

• For H-atom $\mathrm{E_n} = -2.18 \times 10^{-18}\left(\frac{1}{\mathrm{n}^2}\right) \Rightarrow -2.18 \times 10^{-18} \times \frac{1}{2^2}$

$= -0.545 \times 10^{-18}\mathrm{J}$

(57) Answer : (1)
Solution:
K.E of electron $= 13.6 \times \frac{\mathrm{Z}^2}{\mathrm{n}^2} \Rightarrow \mathrm{K.\ E} \propto \mathrm{n}^{-2}$

Angular momentum $= mvr = \frac{nh}{2\pi}$

i.e. Angular momentum $\propto$ n

Radius of nth orbit $= 0.529 \times \frac{n^2}{Z}$

i.e. $r_n \propto n^2$

Velocity of electron in n^{th} orbit $= 2.188 \times \frac{\mathrm{Z}}{\mathrm{n}} \times 10^6 \mathrm{m/s}$

i.e. $v_n \propto n^{-1}$

(58) Answer : (4)
Solution:

$\Delta v = \frac{2 \times 30}{100} = \frac{60}{100} = 0.6$ m/s

$\Delta x \times \Delta v = \frac{\mathrm{h}}{4\pi \mathrm{m}}$

$\Delta x = \frac{6.62 \times 10^{-34}}{4 \times 3.14 \times 50 \times 10^{-3} \times 0.6}$

$\Delta x = 1.756 \times 10^{-33}$ m

(59) Answer : (2)
Solution:

n_1	n_2		Series
1	2, 3…	$\rightarrow$	Lyman
4	5, 6…	$\rightarrow$	Brackett
3	4, 5…	$\rightarrow$	Paschen
5	6, 7…	$\rightarrow$	Pfund

(60) Answer : (4)

Aakash Medical|IIT-JEE|Foundations

Solution:
For a given l value, m can be
–l, –(l – 1) ……0, 1 ….. (l – 1), l
–l to +l including zero.

(61) Answer : (2)
Solution:

$$\frac{r_3 He^+}{r_2 Li^{2+}} = \frac{a_0 \frac{n^2}{z_{He}}}{a_0 \frac{n^2}{Z_{Li}}} = \frac{\frac{3^2}{2}}{\frac{2^2}{3}} = \frac{3^3}{2^3} = \frac{27}{8}$$

(62) Answer : (2)
Solution:

$\frac{1}{\lambda} = R\left[\frac{1}{2^2} - \frac{1}{n^2}\right]$

$= R\left[\frac{n^2-4}{4n^2}\right]$

$\lambda = \frac{4}{R}\left[\frac{n^2}{n^2-4}\right]$

$\lambda = K \times \frac{n^2}{n^2-4}$

so $K = \frac{4}{R}$

(63) Answer : (3)
Solution:
Splitting of spectral lines in the presence of electrical field is called Stark effect.

(64) Answer : (2)
Solution:

Orbital	Number of radial nodes (n – l–1)
2*p*	2 – 1 – 1 = 0
3*p*	3 – 1 – 1 = 1
3*s*	3 – 0 – 1 = 2
4*s*	4 – 0 – 1 = 3

(65) Answer : (2)
Hint:
According to Pauli's exclusion principle; no two electrons in an atom can have the same set of four quantum numbers.
Solution:
It follows only Pauli's exclusion principle. It violates Hund's rule of maximum multiplicity and Aufbau principle.

(66) Answer : (1)
Solution:
The lowest value of n that allows g orbitals to exist is 5.

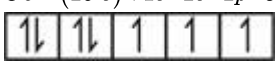
(67) Answer : (3)
Solution:

$\lambda_1 = \frac{h}{\sqrt{2mKE}}$; $\lambda_2 = \frac{h}{\sqrt{2m \times 4KE}} = \frac{h}{\sqrt{2mKE}} \times \frac{1}{2}$

$= \frac{1}{2}\lambda_1$

(68) Answer : (1)
Solution:

$Co^{2+}\left(25\ e\right) : 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^7$



Number of unpaired electrons = 3

(69) Answer : (3)
Solution:
E.C. of Fe^{3+} is $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5$
11 electrons can have m = 0
(For p_Z, m = 0 & for 's' subshell/orbital m = 0)

(70) Answer : (2)
Hint:
Exchange energy of electrons is defined for degenerate orbitals.
Solution:

Exchange energy of electrons is defined as energy released when an electron exchanges its position with electron having same spin present in degenerate orbitals.

(71) **Answer :** (2)
Solution:
Modern periodic Law: The physical and chemical properties of the elements are periodic functions of their atomic numbers.

(72) **Answer :** (4)
Hint:
Chromium and copper have exceptional electronic configuration.
Solution:
$_{29}Cu = 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^1$

$_{24}Cr = 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^1$

$_{25}Mn = 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^2$

$_{26}Fe = 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^6\ 4s^2$

Element	Unpaired electron(s)
Cu	1
Mn	5
Cr	6
Fe	4

(73) **Answer :** (4)
Solution:
EKa – Aluminium is called 'Ga'
EKa – Silicon is called 'Ge'

(74) **Answer :** (3)
Hint:
Atomic weight of middle element is about half of the atomic weights of other two elements in Dobereiner's Triad if elements are arranged in increasing atomic weights.
Solution:
Some example of triads are as
• Li, Na, K
• Ca, Sr, Ba
• Cl, Br, I

Aakash Medical|IIT-JEE|Foundations

(75) **Answer :** (1)
Solution:

Atomic Number	IUPAC official Name
102	Nobelium
104	Rutherfordium
106	Seaborgium
108	Hassium

(76) **Answer :** (4)
Solution:
Element with atomic number 33 is **Arsenic**.
[Ar] $3d^{10}\ 4s^2\ 4p^3$
period number = maximum value of principal quantum number = 4
Group number = 10 + 5 = 15

(77) **Answer :** (3)
Hint:
Z = 60
Eletronic configuration : $[_{54}Xe]4f^4\ 5d^0\ 6s^2$
Solution:
Z = 60 is Nd (Neodymium)
• Since last electron enters *f* subshell element belongs to *f* block.
• Highest value of n = 6 ⇒ 6th period.
• *f* block elements belong to 3rd group.

(78) **Answer :** (3)
Solution:

Element	Atomic Number
Thorium	90
Plutonium	94
Curium	96
Berkelium	97

(79) **Answer :** (4)
Solution:
16th group elements are called chalcogens (ore-forming).

(80) **Answer :** (3)
Solution:
For isoelectronic species, ionic radii decreases as the atomic number increases.

(81) **Answer :** (2)
Solution:
Co (Cobalt) : 125 pm
Ni (Nickel) : 124 pm

(82) **Answer :** (1)
Solution:
$O(g) + e^- \longrightarrow O^-$, $\Delta H° = -142$ kJ mol^{-1}
But O^- resists the addition of another electron and $O^-(g) + e^- \longrightarrow O^{2-}(g)$, $\Delta H° = 844$ kJ mol^{-1}

(83) **Answer :** (2)
Solution:
Correct order of 1st Ionization energy is
B < Be < C < O < N

(84) **Answer :** (1)
Solution:

Element	Δ_{eg} H(kJ mol^{-1})
Br	– 325
F	– 328
S	– 200
Se	– 195

Aakash Medical|IIT-JEE|Foundations

(85) **Answer :** (1)
Solution:
Chlorine has highest negative electrons gain enthalpy which belong to 3rd period group 17 in periodic table.

(86) **Answer :** (4)
Solution:

Ion	No. of electrons
Nickel	Transition metal
Chlorine	Non-metal
Germanium	Metalloid
Samarium	Lanthanoid

(87) **Answer :** (1)
Solution:

Element	Electronegativity value
Si	1.8
C	2.5
N	3.0
O	3.5

(88) **Answer :** (4)
Hint:
% of ionic character = $16(X_A - X_B) + 3.5(X_A - X_B)^2$
Solution:

% of ionic character
$= 16(4 - 3) + 3.5(4 - 3)^2$
$= 16 + 3.5 = 19.5\%$

(89) Answer : (3)
Hint:
Generally metallic oxides are basic and non-metallic oxides are acidic in nature.
Solution:
NO → Neutral
CO_2 → Acidic
ZnO → Amphoteric
CaO → Basic

(90) Answer : (4)
Solution:
The oxidation state of Al is +3 and covalency is 6.
$x - 1 + 0 \times 5 = +2$
$x = +3$

[1 Mark]

BOTANY

(91) Answer : (3)
Hint:
Cell division will occur for eight times.
Solution:
Number of generations $= \frac{160}{20} = 8$
Number of cells = $2^8 \times 10^3$ per ml.
$= 256 \times 10^3$ cells/ml

(92) Answer : (2)
Solution:
Syncytium represents multinucleate condition.
Failure of cytokinesis after karyokinesis leads to formation of syncytium.

(93) Answer : (1)
Solution:
During mitotic anaphase, the centromere splits, chromatids separate and move to the opposite poles.

(94) Answer : (2)
Hint:
DNA replication takes place only once during meiosis.
Solution:
Meiosis involves two sequential cycles of nuclear and cell division but only a single cycle of DNA replication.

(95) Answer : (4)
Solution:
Gap phase 1 of interphase does not involve synthesis of DNA.

(96) Answer : (3)
Solution:
Duration of cell cycle in yeast is generally 90 minutes.

(97) Answer : (2)
Solution:
Attachment of spindle fibres to the kinetochores occurs in the metaphase stage.
Chromatin condensation is an event of prophase.
Exchange of genetic material between homologous chromosomes occurs in meiosis I.

(98) Answer : (2)
Solution:
The given figure represents telophase and it is end of karyokinesis.

(99) Answer : (1)
Solution:
Interphase lasts for more than 95% duration of cell cycle while interkinesis is a short-lived phase.

(100) Answer : (2)
Solution:

Aakash Medical|IIT-JEE|Foundations

Double metaphasic plate is formed during metaphase I of meiosis-I

(101) Answer : (4)
Solution:
In zygotene stage, homologous chromosomes pair with each other. If 40 chromosomes are in G_1 phase, then 20 bivalents will be formed in zygotene.

(102) Answer : (2)
Solution:
Phragmoplast is formed by Golgi complex and grows centrifugally to form cell plate.

(103) Answer : (4)
Hint:
Metaphasic plate is formed on the equatorial plane.
Solution:
The metaphase plate is an imaginary plane where centromere of chromosomes align themselves.

(104) Answer : (2)
Hint:
It is the phase of actual cell division.
Solution:
M-phase is the most dramatic phase of the cell cycle.

(105) Answer : (1)
Solution:
In early prophase, chromosomes appear as ball of wool. This stage of chromosome is known as spireme stage.

(106) Answer : (4)
Solution:
Cell plate appears at centre of cell and then progresses towards periphery.
Duplicated cell organelles get distributed between daughter cells during cytokinesis. Plants have plastids but they do not form furrow.

(107) Answer : (3)
Solution:
Amount of DNA doubles after S phase but number of chromosomes remains same.

(108) Answer : (2)
Hint:
It is preparatory phase of cell cycle.
Solution:
Interphase is preparatory phase of cell cycle known as resting phase because cell duplicates its organelles and replicates its DNA during this phase, but actual division takes place during M-phase.

(109) Answer : (4)
Solution:
Mitosis or M phase is the phase of actual cell division. The daughter cells are relatively smaller than parent cell and small size have high surface area to volume ratio.

(110) Answer : (1)
Solution:
Cell cycle is the period between the formation of new daughter cells and their further division. Interphase is a phase between two successive M phase, where the cell prepares itself for cell division.

(111) Answer : (2)
Solution:
Nucleolus disappears during prophase and reappears during telophase. In haploid organisms, the gametes are formed by mitosis.

(112) Answer : (4)
Solution:
Replacement of old dead and worn out cells is a significance of mitosis

(113) Answer : (1)
Solution:
Nuclear division and cytokinesis occur twice in meiosis

(114) Answer : (2)
Hint:
Anaphase I does not show centromere splitting.
Solution:
During anaphase I, the homologous chromosomes separate, while sister chromatids remain associated at their centromeres.

(115) Answer : (4)

Solution:
In diakinesis, terminalisation of chiasmata occurs. Also during diakinesis, the chromosomes are fully condensed and the meiotic spindle is assembled.

(116) Answer : (4)
Solution:
To form 256 cells 8 mitotic divisions are required but cell divide 255 times.

(117) Answer : (2)
Solution:
Recombinase enzyme catalyses crossing over which occurs during pachytene.

(118) Answer : (1)
Hint:
A tetrad comprises of 2 centromeres.
Solution:
During pachytene stage, the four chromatids of each bivalent become distinct and clearly appear as a tetrad.

(119) Answer : (4)
Solution:
The correct sequence is:
c. Complete disappearance of perinuclear space.
e. Attachment of spindle fibres to the sites found on the primary constriction.
f. Splitting of primary constriction.
a. Formation of the site of active ribosomal RNA synthesis.

(120) Answer : (4)
Solution:
In oocytes of some vertebrates, diplotene can lasts for months or years.

(121) Answer : (2)
Hint:
Chromosomal fibres dissolve and shorten during anaphase that makes the chromosomes to move towards the poles.
Solution:
Interzonal fibres are formed between the centromeres of the sister chromatids. During anaphase, these fibres elongate and move the centromeres apart from each other.

(122) Answer : (4)
Solution:
'Rice', 'Monocot', and 'Plant' are all taxa – but you know that rice is a monocot and monocots are plants. Therefore, 'plants', 'monocot', and 'rice' represent taxa at different levels.

(123) Answer : (4)
Solution:
Tiger, leopards and lions are not in the same biological family as dogs. Highest common category among mango and wheat is kingdom plantae. Potato and brinjal belong to family solanaceae.

(124) Answer : (2)
Solution:
Taxonomy is a process of classification that is based on the characteristics of all living organisms that can be classified into different taxa.
Systematics takes into account evolutionary relationships between organisms.
Classification is a process by which anything is grouped into convenient categories based on some easily observable characters.
Taxonomic categories are distinct biological entities and not merely morphological aggregates.

(125) Answer : (1)
Solution:
Both the words in a biological name when hand written are underlined separately or printed in italics to indicate their Latin origin.
The first word denoting the genus starts with a capital letter while specific epithet starts with small letters.

(126) Answer : (2)
Solution:
The correct decreasing order of taxonomic categories is as follows:
Species → Genus → Family → Order → Class → Division → Kingdom

(127) Answer : (1)
Solution:
Family is placed between order and genus.

(128) Answer : (1)
Hint:
Rudolf Virchow proposed *omnis-cellula-e-cellula.*

Solution:

Ernst Mayr, evolutionary biologist is known as 'The Darwin of the 20th century'.

(129) Answer : (2)
Solution:
Biodiversity is the number and various kinds of organisms found on earth. It stands for the variability found among living organisms inhabiting this world.

(130) Answer : (3)
Solution:
Mangifera, *Solanum* and *Panthera* all are genera.

(131) Answer : (3)
Solution:
Carnivora includes group of organisms belonging to related families.

(132) Answer : (3)
Solution:
Arthropoda is a phylum.

(133) Answer : (2)
Solution:
Two different species cannot breed together to produce fertile offsprings.

(134) Answer : (4)
Hint:
For plants, scientific names are based on agreed principle and criteria provided in ICBN.
Solution:
ICBN stands for International Code for Botanical Nomenclature.

(135) Answer : (3)
Solution:
Datura is placed in family Solanaceae

ZOOLOGY

(136) Answer : (2)
Solution:
All mammals respire by lungs.

(137) Answer : (4)
Solution:
Exchange part is the site of actual diffusion of O_2 and CO_2 between blood and atmospheric air.

(138) Answer : (1)
Solution:
Normal expiration is typically a passive process that happens due to the relaxation of the diaphragm.
Expiration is a passive process as it occurs due to the elastic recoiling of the lungs. Inspiration occurs when diaphragm contracts to increase the overall volume of thoracic cavity.

(139) Answer : (1)
Solution:
Humans have the ability to increase the strength of inspiration and expiration with the help of additional muscles in the abdomen.

(140) Answer : (3)
Solution:
Trachea extends upto the mid-thoracic cavity.
Trachea runs as a straight tube till 5th thoracic vertebra and then divides into right and left primary bronchi.

(141) Answer : (4)
Solution:
Movement of cilia creates a current to transport the materials that come in contact with them.

(142) Answer : (2)
Hint:
Largest class in the animal kingdom
Solution:
Lower invertebrates like sponges, coelenterates, flatworms, *etc.*, exchange O_2 with CO_2 by simple diffusion over their entire body surface.

(143) Answer : (4)
Solution:
Cellular respiration is utilisation of O_2 by cells for catabolic reactions and release of CO_2. Breathing or pulmonary ventilation is a process by which atmospheric air is drawn in and CO_2 rich alveolar air is released out.

(144) Answer : (3)
Hint:
'X' is a cartilaginous box named larynx.
Solution:
Larynx is a cartilaginous box which helps in sound production and hence called the sound box. During swallowing, glottis can be covered by a thin elastic cartilaginous flap, called epiglottis, to prevent the entry of food into the larynx.

(145) Answer : (3)
Solution:
Guanine pairs with cytosine and adenine pairs with thymine in a DNA molecule.

(146) Answer : (2)
Solution:
Diffusion of gases is directly proportional to surface area.

(147) Answer : (4)
Solution:
The trachea, primary, secondary and tertiary bronchi and initial bronchioles are supported by incomplete cartilaginous rings.

(148) Answer : (3)
Solution:
The lungs are situated in the thoracic chamber which is anatomically an air-tight chamber.

(149) Answer : (4)
Solution:
Primary structure of proteins tells us the positions of amino acids.

(150) Answer : (3)
Solution:
Glycine has no asymmetric carbon because it has two hydrogen atoms attached to alpha carbon.

(151) Answer : (4)
Solution:
Alanine, valine and glycine are examples of aliphatic amino acids.

(152) Answer : (2)
Hint:
Palmitic acid is a saturated fatty acid.
Solution:
Palmitic acid has 16 carbons including the carboxyl carbon. Therefore, correct representation of palmitic acid is $CH_3 – (CH_2)_{14} – COOH$.

(153) Answer : (4)
Solution:
Steroids are derived lipids.

(154) Answer : (2)
Hint:
Component of lipids.
Solution:
Amino acids are substituted methanes. Amino acids have four substituent groups *i.e*., hydrogen, carboxyl group (–COOH), amino group ($–NH_2$) and a variable group designated as R-group. Glycerol is trihydroxy propane and is a constituent of lipids.

(155) Answer : (2)
Solution:
The most abundant protein present in the animal world is collagen and it is arranged in an irregular pattern in dense irregular connective tissue, that is present beneath the skin.

(156) Answer : (2)
Solution:
In nucleic acid, the bond between the phosphate and hydroxyl group of sugar is an ester bond.

(157) Answer : (2)
Solution:
The essential chemical components of many co-enzymes are vitamins *e.g*., co-enzyme nicotinamide adenine dinucleotide (NAD) and NADP contain the vitamin niacin.

(158) Answer : (4)

Aakash Medical|IIT-JEE|Foundations

Solution:
The tertiary structure of proteins is stabilised by hydrogen bonds, covalent bonds, hydrophobic bonds and ionic bonds. Phosphodiester bonds are found in RNA and DNA.

(159) Answer : (3)
Hint:
Prosthetic group
Solution:
There are three kinds of co-factors–prosthetic groups, co-enzymes and metal ions.
Prosthetic groups are organic compounds and are distinguished from other co-factors in that they are tightly bound to the apoenzyme. For example, in peroxidase and catalase, which catalyse the breakdown of hydrogen peroxide to water and oxygen, haem is the prosthetic group and it is a part of active site of the enzyme. Zinc is a co-factor for the proteolytic enzyme, carboxypeptidase.

(160) Answer : (4)
Hint:
Enzyme-product complex dissociates to form product and enzyme.
Solution:
Each enzyme (E) has a substrate (S) binding site in its molecule so that a highly reactive enzyme-substrate complex (ES) is produced. This complex is short-lived and dissociates into product (P) and unchanged enzyme through an intermediate formation of enzyme-product complex (EP). So, the correct representation of this biochemical reaction is
$E + S \rightarrow ES \rightarrow EP \rightarrow E + P$

(161) Answer : (4)
Solution:
Transferases include the enzymes that catalyse the transfer of a group, other than hydrogen between a pair of substrates.

(162) Answer : (4)
Solution:
Lecithin is an example of a phospholipid.

(163) Answer : (4)
Solution:
Michaelis constant (K_m) refers to substrate concentration at which the reaction velocity reaches half of its maximum velocity.

(164) Answer : (4)
Solution:
Catalytic activity is lost when the co-factor is removed from the enzyme which testifies that they play a crucial role in the catalytic activity of the enzyme.

(165) Answer : (2)
Solution:
A peptide linkage is formed by the removal of a water molecule when two amino acids are joined together.

(166) Answer : (3)
Solution:
Hydrolases are the enzymes which catalyse breakage of bond present between the glycerol and fatty acids in a molecule of triglyceride and their enzyme commission number starts with 3.

(167) Answer : (4)
Hint:
ERV + TV + IRV
Solution:
Vital capacity (VC) is defined as the maximum volume of air a person can breathe in after a forceful expiration or the maximum volume of air a person can breathe out after a forceful inspiration.
Depending upon the age, sex and height of an individual, its value varies from 4000 mL to 4600 mL.
Vital capacity (VC) = ERV + TV + IRV or IC + ERV = 4000 mL to 4600 mL.

(168) Answer : (3)
Solution:
GLUT-4 enables glucose transport into cells. Haemoglobin present in RBCs transports oxygen from lungs to various parts of the body.

(169) Answer : (1)
Solution:
There are 20 types of amino acids which occur in proteins and 4 types of nucleotides are present in RNA.

(170) Answer : (3)
Solution:
2-deoxyribose sugar is present in DNA, where oxygen is absent at 2nd carbon of ribose sugar. Ribose, a pentose sugar, is a monosaccharide.

(171) Answer : (3)
Solution:
Adult human haemoglobin consists of 4 subunits (2α and 2β).

(172) Answer : (1)
Solution:
Total lung capacity (TLC) is the total volume of air accomodated in the lungs at the end of forced inspiration.
Inspiratory capacity is the total volume of air inspired after a normal expiration. Thus, it includes TV (Tidal volume) and IRV (Inspiratory reserve volume).
Vital capacity is the maximum volume of air a person can breathe after a forceful expiration or maximum volume of air a person can expire after a forceful inspiration.
FRC is the volume of air that remains in the lungs after normal expiration.

(173) Answer : (1)
Solution:
Cytosine + Ribose sugar ⇒ Cytidine
A nucleoside consists of a nitrogenous base attached to a sugar molecule.

(174) Answer : (3)
Solution:
Based on the number of amino and carboxyl groups, valine is a neutral and lysine is a basic amino acid.

(175) Answer : (1)
Hint:
Diaphragm is dome-shaped.
Solution:
During the process of normal inspiration, diaphragm and external inter-costal muscles contract which leads to an increase in the volume of thoracic chamber and decrease in intra-pulmonary pressure.
When diaphragm contracts, it becomes flattened in shape.
Abdominal muscles contract during forceful expiration.

(176) Answer : (1)
Solution:
Collagen is the most abundant protein in animal world.

(177) Answer : (2)
Solution:
Each human lung is covered by a double layered pleura: Outer pleura (parietal pleura) and inner pleura (visceral pleura), with pleural fluid between them.

(178) Answer : (3)
Solution:
Tidal volume, IRV and ERV when measured by spirometer reflect a value of 500 mL, 2500-3000 mL and 1000-1100 mL respectively. Residual volume when evaluated by other methods equals 1100-1200 mL.

(179) Answer : (2)
Solution:
On an average, a healthy adult human breathes 12-16 times/minute.

(180) Answer : (3)
Solution:
The conducting part of the respiratory system is not involved in gaseous exchange.

Aakash Medical|IIT-JEE|Foundations